Supplementary Material
for
What depth-resolved transient measurements on a SiC honeycomb solar receiver imply for reduced-order heat transfer closures

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S1 Experimental campaigns on porous and structured volumetric absorbers

The response has largely been to build better absorbers. Fend and co-workers measured thermophysical and heat transfer properties of cordierite, SSiC and CBSiC foams, SiC fibre meshes and catalyst carriers, obtaining efficiencies of 75–92% at 300–550 °C for combined material systems and showing that 45 ppi foams transfer far more heat than 10–20 ppi foams by virtue of specific surface areas of up to 2500 $m^2 m^{-3}$ (Fend et al., 2004a); they also reported double-layer SiC foams reaching 98% efficiency at 672 °C outlet (Fend et al., 2004b). Ávila-Marín and colleagues built and tested twenty-six graded-porosity wire-mesh absorbers against the TSA and SOLAIR baselines (Avila-Marin et al., 2018), and later five periodic and non-periodic SiC morphologies including Voronoi structures that outperformed a conventional SiC honeycomb (Avila-Marin et al., 2022). Zaversky and co-workers optimized single- against multi-layer ceramic foams and concluded that only the first layer's properties matter thermally, with a broad efficiency plateau between 30 and 50 PPI (Zaversky et al., 2018). Wang and co-workers tested four porous SiC absorbers and found that reducing pore diameter from 5.83 to 2.22 mm at fixed porosity raised efficiency by nearly three points (Wang et al., 2021). The pattern continues in the most recent work. Patil and co-workers built and tested a 5 kW reticulated-porous-ceramic cavity receiver for process heat above 1000 °C, comparing SiSiC, alumina and ceria structures across mean pore sizes (Patil et al., 2021), and Broeske and co-workers tested screen-printed ceramic and metallic-foil three-dimensionally shaped absorbers in a sixteen-module matrix at DLR's Synlight facility, reaching air exit temperatures to 735 °C and thermal efficiencies up to 91.8% at 650 °C against 84.6 to 85.3% for the 80 cpsi HiTRec SiSiC honeycomb benchmark (Broeske et al., 2023). D'Souza and co-workers took the complementary route of a calorimetric test bench, measuring the heat transfer to pressurised gas in absorbers of varying rectangular channel geometry under a high-flux solar simulator (D'Souza et al., 2025). The recurring output of this family of work is a design recommendation — a porosity, a cell density, a layering, a channel aspect ratio — obtained from steady-state efficiency measurements.

The recurring output of these campaigns is a design recommendation — porosity, cell density, layering, channel aspect ratio — inferred from steady-state efficiency and outlet-temperature measurements. Section 5.1 of the manuscript quantifies what such a dataset constrains.

S2 Uncertainty propagation

Three kinds of uncertainty are propagated and reported separately, because they behave differently.

Random and calibration error is propagated by Monte Carlo over 4000 realizations. Reported ± values are Monte Carlo standard deviations and bracketed ranges are 95% percentile intervals. The type N

tolerance of IEC 60584 class 1 — the greater of 1.5 K and 0.004|t| — is a maximum permissible error rather than a standard deviation, so it is read as a 95% bound and halved to a 1σ term. That term is a per-sensor calibration offset, so it is drawn once per sensor per realization and shared across all runs, scaled by each run's temperature-dependent tolerance, with an independent 0.5 K steady-window drift term per run. Viscosity and conductivity carry 2% and 1% perturbations. The abscissa of the transient identification, x = ε ṁ c_p, is rebuilt in every realization from that realization's perturbed flow and temperatures, with the controller factor shared between the steady frame and the abscissa because it is one physical error.

The controller model is two-term. The four Aalborg GFC17 units carry roughly 25, 33, 28 and 14% of the total flow, and two independent pieces of information constrain their error. The manufacturer specification is ±1.0% of full scale over the whole 0 to 100% range with ±0.5% full-scale repeatability — an additive bound in absolute flow, not a percent of reading — and the factory calibration is gas-specific. Because the receiver runs air rather than the calibration gas, each unit was calibrated in place against a bubble flowmeter; that substitution removes the gas-conversion factor, which is a percent-of-reading systematic, and replaces it with the residual uncertainty of the measured characteristic, which is also proportional to reading. What it cannot remove is the unit's own zero, linearity and repeatability floor, which remains additive. Both terms are therefore carried,

$$\sigma_u = [(a_{\mathrm{FS}}F)^2 + (b_{\mathrm{rel}}\, q_u)^2]^{1/2}, \qquad a_{\mathrm{FS}} = 0.0025,\ b_{\mathrm{rel}} = 0.025,\ F = 10\ \mathrm{sL\,min^{-1}},$$

with the additive coefficient the full-scale repeatability read as a 95% bound and the proportional coefficient the calibration residual. One standard-normal draw per unit per realization is combined with each record's own measured shares, so a single physical unit error reaches every steady and transient record at the magnitude its own reading implies.

The two terms have opposite flow dependence, and in this campaign neither is negligible: per-unit readings run 0.69 to 5.72 sL min^{-1}, that is 7% to 100% of full scale, and the additive term exceeds the proportional one for 48% of the per-unit readings. On the summed flow the relative 1σ falls from 3.4% at the lowest campaign flow to 2.6% at the highest for fully correlated units, and from 1.7% to 1.3% for independent ones. A purely proportional model would give a flow-independent 2.5% and would therefore, by construction, be unable to reach a fitted exponent at all; the correct model is predominantly systematic, so it displaces a prefactor, but retains a weak gradient that does reach an exponent. Because the manufacturer states no correlation between units, every instrumental term is computed in both limits: they agree to within 0.0007 on every exponent and within 0.5 J K^{-1} on every capacitance, so the larger is quoted and both are archived in `uncertainty.csv` under `rho_mfc`.

Where a quantity is a fitted slope, its regression standard error is reported alongside and labelled as such, because the two measure different things: the Monte Carlo term is what the instruments contribute and the regression term is the scatter of the runs about the fitted law. For the pooled Nusselt correlation the regression term is larger by a factor of 22.7 and for the profile-corrected transfer-unit count by 13.1, so in both cases it is the one to read as the uncertainty of the correlation; for the grouped Nusselt correlation, which is much better determined, the ratio is only 3.4 and the instrumental term is not negligible.

Systematic bias in the outlet gas probe is propagated as a declared band, because it is not a calibration error and is much larger than one. T3 sits in the exit plenum at a convective–radiative equilibrium between the gas and the surrounding hardware and reads neither bulk gas nor wall temperature; the bias is of order tens of kelvin and is not characterized, a difficulty identified as a principal source of the model–experiment gap in this field (Capuano et al., 2016) and demonstrated for thermocouples in monolith channels specifically (Hayes and Cornejo, 2021). Since ε, NTU, h_{app}, Nu,

ε*, η and — through ε — C_{eff} and K_{loss} all derive from T3, each is reported with a δT3 = ±25 K band alongside its Monte Carlo interval. The band is carried through the identification as well as through the steady groups, by recomputing the abscissa at each offset with the same estimator the archived value uses, so that the zero-offset case reproduces the archived point estimate exactly:

quantity	δT3 = −25 K	δT3 = 0	δT3 = +25 K	band
C_{eff}, cooling matched-ε (primary) [J K^{-1}]	271.3	275.5	280.1	271.3 – 280.1
K_{loss}, cooling matched-ε (primary) [W K^{-1}]	0.0843	0.0796	0.0752	0.0752 – 0.0843
C_{eff}, heating deep probes [J K^{-1}]	262.0	280.6	299.2	262.0 – 299.2
K_{loss}, heating deep probes [W K^{-1}]	0.1072	0.1136	0.1200	0.1072 – 0.1200

The identified capacitance is thus the quantity least sensitive to the outlet-probe systematic — the primary determination moves by 1.6% over the whole band, against the 11% spread among estimators and the ±37 J K^{-1} Monte Carlo standard deviation — while the loss bracket widens from 0.080–0.114 to 0.075–0.120 W K^{-1}. The structural result of section 4.2 is deliberately constructed to survive the band.

Estimator choice is propagated for the inversion threshold, where it dominates: section 4.1 reports ε* located both by interpolation between the bracketing runs and by a global linear regression, and carries the difference as a systematic.

S3 Conditionality of the heating identification

The heating approach to steady state decays with the same eigenvalue in principle, and $log(T_{ss} - T)$ is regressed over a deficit window in $u = (T_{ss} - T)/(T_{ss} - T_0)$. Two choices are unresolvable within this dataset. Sensor selection. With the lamps on, the front probes are held by a local balance between absorbed flux and aperture reradiation with its own short time constant, so their approach to steady state is bi-exponential and does not carry the assembly-scale mode. Using all six receiver probes gives C_{eff} = 121.6 J K^{-1}; using the three deep and outlet probes gives 280.6 J K^{-1}, a ratio of 2.31.

Fit window. Sweeping the deficit window moves the identified capacitance monotonically while the coefficient of determination improves, so goodness of fit does not select the window.

deficit window u	C_{eff} [J K^{-1}]	K_{loss} [W K^{-1}]	r^2
(0.05, 0.35)	294.7	0.1273	0.8407
(0.07, 0.45)	280.6	0.1136	0.9034
(0.10, 0.50)	263.7	0.0999	0.9359
(0.15, 0.60)	227.9	0.0739	0.9621
(0.20, 0.70)	199.5	0.0564	0.9665

Sensor selection: all six receiver probes give 121.6 J K^{-1}, the three deep and outlet probes 280.6 J K^{-1}, a ratio of 2.31.

S4 Wall-profile reconstruction and the fixed-conductance falsification

The length-averaged wall temperature and the profile-corrected transfer-unit count both use a piecewise-linear axial wall profile through the probes at 11, 58 and 107 mm over a 137 mm receiver. The front 11 mm and rear 30 mm are unmeasured. Four admissible extrapolation combinations give exponents within 0.012 of one another; a full linear continuation of the rear gradient is inadmissible, because for five of the fifteen runs it places the exit wall below the measured outlet gas temperature, which no single-stream model can produce.

Rear / front extrapolation	exponent	s.e.	runs	infeasible
constant / constant	+0.3408	0.0404	15	none
constant / linear	+0.3429	0.0408	15	none
half / constant	+0.3307	0.0409	15	none
half / linear	+0.3329	0.0413	15	none
linear / constant	—	—	10	E71, E75, E76, E80, E81
linear / linear	—	—	10	E71, E75, E76, E80, E81

The primary value is the constant/constant case, +0.3408 ± 0.0404, against the single-stream requirement of -1.00. Under the common-slope grouping used for the Nusselt correlation the exponent is +0.3558 ± 0.0099 with $r^2 = 0.9922$.

The fixed-conductance falsification of section 5.2 fits a piecewise-linear non-negative h(z) by multi-start least squares in log h and asks whether one flow-independent profile reproduces the fifteen measured outlet temperatures. It does not, in any family or at any node count, and the discriminating statistic is the ordering of the residual in flow rather than its magnitude. For the shared conductance profile with five nodes, the residual runs from +62 to +95 K at the lowest flow of each configuration to −27 to −85 K at the highest.

Family	nodes	RMS residual [K]	max abs [K]	r vs $\ln Re$	slope [K per e-fold]
shared dimensional $h(z)$	2	64.2	111.3	-0.943	-148
shared dimensional $h(z)$	3	55.4	96.5	-0.946	-128
shared dimensional $h(z)$	5	54.2	95.0	-0.947	-126
shared dimensional $h(z)$	7	54.2	95.0	-0.947	-126
shared $Nu(z)$	3	55.8	95.7	-0.946	-132
shared $Nu(z)$	5	54.6	94.1	-0.948	-129

per-configuration, 256 kW m^{-2}	5	36.0	56.9	-0.996	-90
per-configuration, 304 kW m^{-2}	5	67.7	97.3	-0.982	-128
per-configuration, 456 kW m^{-2}	5	52.1	79.1	-0.997	-197

S5 Auxiliary dimensionless groups, per run

The four auxiliary groups close off the local explanations for the assembly-scale exchange deficit of section 4.2: the web Biot number, the Graetz number with its reciprocal $x^* = 1/Gz_L$ (the dimensionless axial coordinate of the thermal-entrance problem at the channel exit), the axial Péclet group, and the radiation–conduction number.

Run	G_0 [kW m^{-2}]	q [sL min^{-1}]	Re_{nom}	Pr	Gz_L	$x^* = 1/Gz_L$	$Re\,Pr\,L/D_h$	Bi	N_{rc}
E67	456	15.28	72.2	0.6994	0.553	1.81	4610	2.53e-05	4.32
E68	456	12.50	58.6	0.6996	0.449	2.23	3744	1.88e-05	4.80
E69	456	10.50	49.4	0.6995	0.378	2.64	3155	1.48e-05	5.01
E70	456	9.11	42.5	0.6998	0.326	3.07	2717	1.19e-05	5.55
E71	456	7.13	33.8	0.6993	0.259	3.86	2159	8.38e-06	5.57
E72	304	18.32	93.6	0.6980	0.715	1.40	5964	3.38e-05	2.69
E73	304	13.17	65.2	0.6983	0.499	2.01	4160	2.17e-05	3.40
E74	304	9.03	44.6	0.6984	0.341	2.93	2845	1.25e-05	3.92
E75	304	6.95	34.7	0.6982	0.266	3.76	2216	8.53e-06	4.07
E76	304	4.53	23.2	0.6980	0.178	5.63	1481	4.80e-06	4.04
E77	256	13.85	78.5	0.6985	0.600	1.67	5008	2.60e-05	1.53
E78	256	10.01	55.3	0.6982	0.422	2.37	3524	1.68e-05	1.87
E79	256	8.04	44.2	0.6981	0.338	2.96	2818	1.22e-05	2.03
E80	256	6.61	36.3	0.6981	0.277	3.60	2314	9.31e-06	2.15
E81	256	4.53	25.1	0.6982	0.192	5.22	1599	5.54e-06	2.25

S6 Delivered aperture irradiance

The aperture irradiance is determined two ways.

The first is radiometric flux mapping, using the three-step characterization procedure established for this simulator (Abdellateif et al., 2023; Haseler et al., 2021): an alumina-coated Lambertian target on a precision three-axis slider, a Gardon-type circular foil radiometer traversed across the focal region on a graded grid with NIST-traceable calibration and calorimetric correction for spectral mismatch, and a parallel high-dynamic-range CCD reconstruction whose grayscale-to-flux transfer fits a power law with $a = 1117.5$ W m^{-2} and exponent 0.903. Gaussian fits to the resulting profiles validate the focal distribution with normalized RMSE of about 0.07 and 0.12 on the two axes (Haseler et al., 2021). Integrating the calibrated map over the receiver face gives nominal values of 456, 304 and 256 kW m^{-2} for the three configurations.

The second is an apparent steady energy closure, not model-free: it takes K_loss from the lumped identification of section 4.4 and omits storage by assuming the assembly is stationary at run end. Neither assumption is safe: insulation probe T2 rises monotonically as flow falls and run duration lengthens, from 308 to 356 K across the campaign, with decay far slower than any run window, so insulation is still charging when runs end, leaving a residual storage term of unknown sign. Nor need a K_loss from a cooling secant equal the steady loss under irradiation, since part of the path is radiative and insulation temperature differs between states. At steady state:

$$f = \frac{Q_{\mathrm{gas}} + K_{\mathrm{loss}}(\bar{T}_w - T_{\mathrm{amb}})}{G_0 A_{\mathrm{frt}}}$$

which should equal unity if the radiometric flux is correct. Evaluated at both ends of the identified K_loss bracket and averaged per configuration, it does not: f runs 0.988–1.133 at 456 kW m^{-2}, 1.148–1.323 at 304 and 0.783–0.915 at 256, giving closure estimates of 450–517, 349–402 and 201–234 kW m^{-2}, with reported spans 450–517, 304–402 and 200–256 kW m^{-2}.

We report both estimates, with every power-normalized quantity spanning them. This is not a bounding interval: closure omits storage, inherits a lumped loss model, and carries T3 bias directly into Q_gas, so neither determination is an established bound and the disagreement cannot be assigned to one cause. Apparent efficiency on the nominal basis ranges 0.252–1.223, exceeding unity in the 304 kW m^{-2} group.

One contributor is geometric. The Gardon gauge reports flux over its 12.6 mm-radius exposed area (Haseler et al., 2021), 499 mm^2 against the receiver's 361 mm^2 face, a ratio of 1.38. In a focused Gaussian spot the gauge averages a larger, more peripheral region than the receiver intercepts, under-reading the face flux: the face-to-gauge ratio for a centred Gaussian is 1.28, 1.17, 1.11 and 1.04 for $1/e^2$ radii of 10, 14, 18 and 30 mm, bracketing the 1.13 required at 456 kW m^{-2} and covering roughly half the 1.32 required at 304. This is structural to characterizing a 19 mm receiver with a 25 mm gauge, not execution error, but it does not explain 256 kW m^{-2}, where closure sits 8–16% below nominal and a broader spot pushes the factor toward unity, so aperture spillage dominates oppositely. Comparably, optical modelling of a directly irradiated reactor on the same simulator found only 1–6% of aperture-incident power reaching the catalytic bed (Abdellateif et al., 2023). Supplementary section S6 gives the gauge-averaging calculation and remaining instrument contributions.

Outlet-probe bias alone cannot explain the discrepancy: the uniform T3 shift bringing closure to f = 1 is −75 K at 456 kW m^{-2}, −126 K at 304 and +75 K at 256 — sign-varying and varying 250 K within a configuration across flow, so no uniform correction reconciles the two determinations. A flow-dependent energy-budget term is required, without identifying which: bypass around the monolith periphery and flow maldistribution among channel groups are candidates consistent with section 4.2, but non-stationarity or flow-dependent probe response are not excluded. Convergence with section 5.2 is a consistency, not independent confirmation.

No temperature-based result depends on this choice: effectiveness, transfer units, the Nusselt correlation, the inversion criterion, disequilibrium indices, eigenvalue identification and similarity collapse are all ratios of measured temperatures and flows. Only efficiency and delivered-power-per-unit-mass figures shift, reported as spans rather than intervals containing the true value.

S7 What a flow-dependent split requires

That prescription has been carried out elsewhere, which makes the recommendation concrete rather than aspirational. Nagarajan, Ström and Sjöblom build a reduced-order pseudochannel model of a

catalytic converter precisely because single-channel models cannot predict accurately under flow maldistribution in realistic geometries (Chanda Nagarajan et al., 2025) — the remedy being a few representative channels carrying different flows rather than one channel with a better closure. The same group whose channel-scale Nusselt work anchors section 4.2 has developed flow-distribution models for dual-cell-density monoliths (Reinao and Cornejo, 2023) and dual-zone packed beds (Díaz et al., 2025). Within the solar field the step has been taken at panel and module scale: Schwager and co-workers resolve the mass-flow distribution among the parallel tubes of a molten-salt receiver panel under flux gradients (Schwager et al., 2024), and the multi-zone unsteady model of Kumar and co-workers resolves exchange between absorbers, casing and return air rather than treating the receiver as one stream (Kumar et al., 2022).

Those two flow-distribution studies do more than set a precedent: they identify what a flow-dependent split requires. Reinao and Cornejo model a dual-cell-density monolith as two concentric regions of differing apparent permeability with a Darcy momentum sink, isothermal, and find the flow fraction through each region depends only on relative permeability and core size, not on flow rate, across several flow rates (Reinao and Cornejo, 2023). That invariance is exact: when every parallel path has a resistance linear in velocity and shares one viscosity, flow rate cancels from the equal-pressure-drop condition. Díaz and co-workers repeat the exercise for a dual-zone packed bed, where the Ergun equation adds an inertial term, and there the split does depend on Reynolds number — strongly sensitive to the resistance contrast at low Re where the viscous term dominates, converging as inertia takes over (Díaz et al., 2025). A split that changes with flow rate therefore cannot arise in parallel linear-viscous paths of common viscosity; it requires an inertial contribution or a path-to-path viscosity difference. This receiver's participation does change with flow, as Re_nom^+0.341, and both admissible causes are present: the heated absorber carries a non-uniform viscosity field (Wang et al., 2021), and the viscous-to-inertial transition is the same axis as the linear-versus-quadratic stability criterion of section 4.7 (Becker et al., 2006; Kribus et al., 1996). The direction matches too, the packed-bed split growing more uniform as flow rises.

A third agent gives the same structure. In Schwager's molten-salt panels the destabilizing force is buoyancy rather than inertia or viscosity — hotter, less dense salt gains flow in an upward-flowing panel but loses it in a downward-flowing one, where the effect is self-reinforcing and can drive tubes to zero flow — and what arrests it is friction, so reducing overall flow intensifies the maldistribution: below 20% of nominal flow mild flux spreads suffice to trigger reversal, and at 10% no spread is needed (Schwager et al., 2024). That mechanism is not this receiver's, but in all three cases a driving force for maldistribution competes with a frictional resistance whose relative weight grows with flow, so the split is most differentiated at low flow and most uniform at high flow. The flow-rate dependence of participation is therefore a general property of heated parallel-path arrays rather than an artefact of one geometry.

S8 Pressure drop as installed

The measurement is judged against the decomposition standard for a flow-through monolith, that is, for a substrate in which every channel is open to through flow rather than alternately plugged: the total drop between the cavity and the exit plenum is the sum of a face contraction, core friction, a hydrodynamic entrance contribution, thermal acceleration and a face expansion (Kays, 2018; Shah and London, 1978). Core friction uses the square-duct Poiseuille number of 56.91 with all metered flow through the 100 channels, but is integrated along the channel with temperature-dependent properties instead of being evaluated once at $\bar{T}_g$. At constant channel mass flux and pressure the velocity rises in proportion to the gas temperature while the viscosity of air rises roughly as T^0.7, so the friction integrand scales as T^1.7 and is weighted towards the hot end of the channel; the gas profile is reconstructed by integrating the single-stream energy equation against the measured piecewise-linear

wall profile with the transfer-unit count fixed per run so that the exit value reproduces the measured T3, which raises the length-mean gas temperature from 411–539 K to 441–612 K. The entrance contribution uses the square-duct incremental pressure-drop number $K(\infty) = 1.55$ (Shah and London, 1978); Shah's apparent friction factor (Shah, 1978) evaluated with the same constants stays within 1.0 to 3.9% of the fully developed value over the campaign ($x^+ = 0.69$ to 2.8 against a laminar entry length of $x^+ \approx 0.09$ and channel Reynolds numbers of 33 to 133), so the channels are hydrodynamically developed. The face terms use the sharp-edged contraction and full dissipation of the exit velocity head appropriate to the cavity and plenum that bound the monolith; together with thermal acceleration they contribute 1.2 to 4.9% of the predicted drop, which is consistent with the monolith-reactor literature in finding inlet and outlet minor losses negligible for a single substrate of this slenderness and significant only for short or closely coupled ones (Cornejo, 2021; Cornejo et al., 2020).

The resulting prediction is 0.24 to 1.29 mbar, a uniform 1.19 to 1.30 times the isothermal single-duct estimate it replaces, and 95 to 99% of it remains core friction, so the correction is a variable-property correction and not a minor-loss one. Against it, the measured signal still exceeds the prediction by 1.4 to 2.1 times at high flow (2.74 against 1.29 mbar at 15.3 sL min^{-1}), falls below it at the low-flow end of the 304 and 256 kW m^{-2} configurations, and goes negative at the lowest flow (−0.02 against 0.24 mbar at 4.5 sL min^{-1}). The ratio rises monotonically with flow in all three configurations but crosses unity only in those two: at 456 kW m^{-2} the measurement exceeds the prediction at every tested flow, the lowest being 1.4 times at 7.1 sL min^{-1}. A negative drop across a resistance in steady forward flow is impossible, so the discrepancy is the signature of a zero offset on a transducer whose ±0.2 mbar accuracy floor, 0.1% of its ±200 mbar span, is the same order as the entire expected monolith drop. The taps evidently span more than the monolith faces, and the second differential channel reads a flat −0.26 to −0.30 mbar in every run and appears unconnected. Figure S1 shows both comparisons.

This matters because pressure drop is how the volumetric-receiver literature diagnoses honeycomb flow behaviour: absorbers with linear velocity dependence tend inherently towards flow instability (Kribus et al., 1996); high-thermal-conductivity materials with a quadratic characteristic are stable (Becker et al., 2006); instability becomes possible when one pressure-drop level maps to more than one outlet temperature, letting a region drift and fail if temperature exceeds material limits (Broeske et al., 2022). Ávila-Marín's review extends this to honeycombs, identifying a critical flux above which highly porous honeycombs become unstable while wire meshes and low-porosity foams stay stable via their quadratic characteristic (Ávila-Marín, 2011); permeability and Forchheimer coefficients from pressure-drop data feed homogeneous equivalent models (Avila-Marin et al., 2019). Recent unheated monolith measurements show a linear Re–pressure-drop relationship throughout, with progressive underestimation as Re rises through entrance/exit effects (Reinao et al., 2025), large enough at channel Re 50–320 to need a spacing-dependent correction factor increased by misalignment (Cornejo, 2024). For straight square channels those homogeneous-equivalent coefficients are not empirical: matching the core-friction term gives a Darcy permeability $K_1 = 2\phi D_h^2/(f_D\,Re) = 4.93 \times 10^{-8}$ m^2, and the Forchheimer coefficient is identically zero in fully developed laminar flow, the face and acceleration terms above supplying the only quadratic content. Our honeycomb therefore sits by construction on the instability-prone, linear side, but our transducer, spanning monolith plus plumbing, cannot separate the absorber's characteristic from comparable entrance, exit and fitting losses. We report this because the fix is inexpensive, and section 5.3 shows this measurement would discriminate between the two candidate mechanisms above.

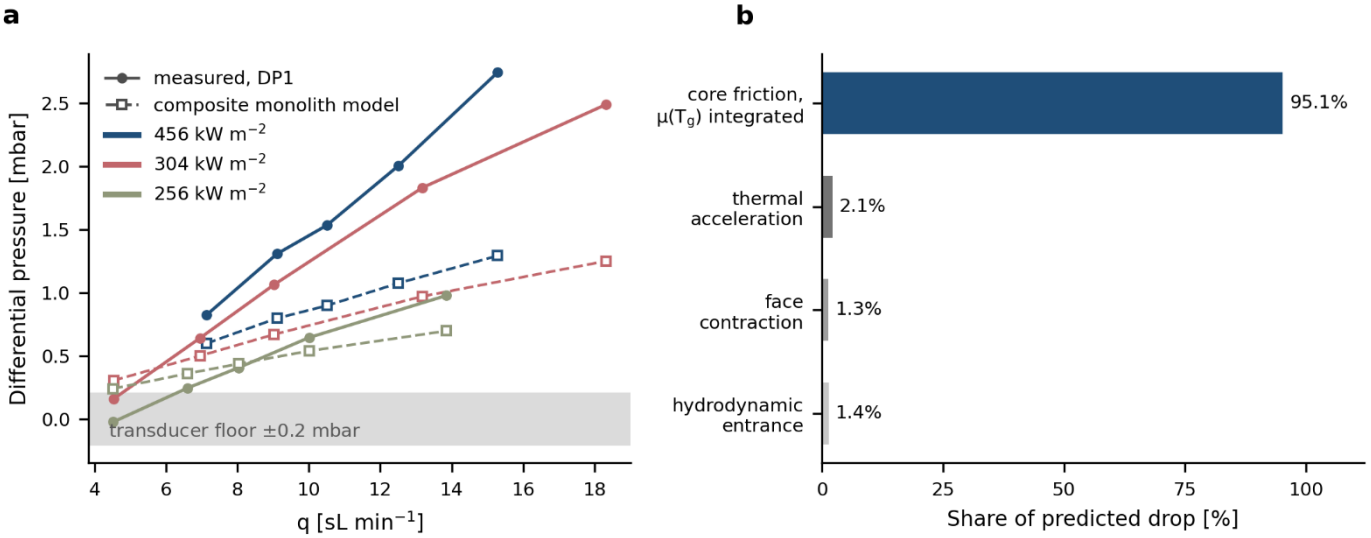


Figure S1 *a) Measured differential pressure DP1 against the composite flow-through monolith prediction for the fifteen heating runs, by irradiance configuration; the shaded band is the ±0.2 mbar accuracy floor of the Keller PD33X; b) Term-by-term composition of the prediction at the highest metered flow of the campaign, where the minor losses are largest.*

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